

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461095.9344 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1601 +/- 0.0084
Transit depth (Rp/Rs)^2	2.56 +/- 0.27 [%]
Orbital Inclination (inc)	89.96 +/- 0.4
Airmass coefficient 1 (a1)	1.599 +/- 0.02
Airmass coefficient 2 (a2)	-0.448 +/- 0.055
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	None
Optimal Aperture	3.65
Optimal Annulus	7.94
Transit Duration (day)	0.10075 +/- 0.00081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1601 ± 0.0084 vs 0.1406 (deviation 2.32σ)
Duration fit vs published	0.10075 d vs 0.0975 d (deviation 4.01σ)
Mid-time O–C	5.8 min
Depth SNR	19.1
Residual scatter	1.24 %

Light curve

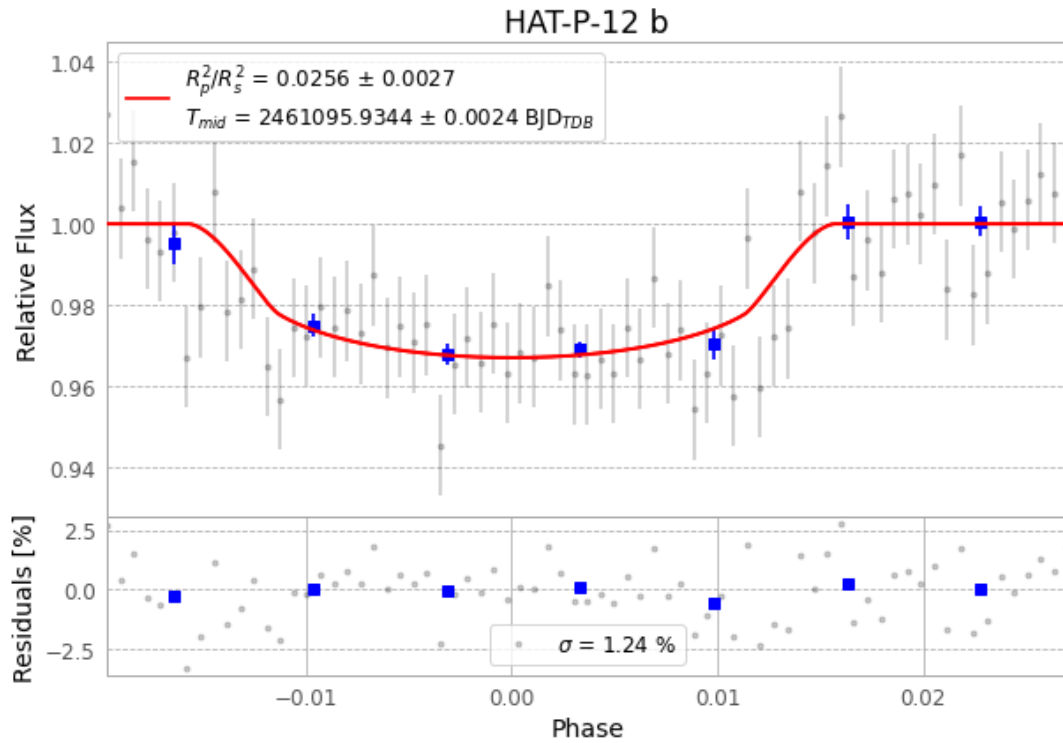


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-02-24.png (SHA-256 baafde1a71aa0468...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 109], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9d67122d1505cb3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

73 FITS frames (48 MB), HATP-12260224084816.FITS → HATP-12260224122416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260224.log (a18ac4c5e091...), FinalLightCurve_HAT-P-12 b_2026-02-24.png (baafde1a71aa...), FinalLightCurve_HAT-P-12 b_2026-02-24.csv (38c5f9f1b04a...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:41Z.